

Daily Geometry Drill #1

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Gradients & Perpendicularity · Distance & Midpoints · Perpendicular Distance · Shoelace Areas · Collinearity.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Gradient, intercept, midpoint, distance — read them straight off. No working shown needed.

- Find the gradient of the line $6x - 3y + 15 = 0$.
- Find the gradient of the line through $(1, 2)$ and $(5, 14)$.
- A line of gradient 4 is perpendicular to a line of gradient m . Write down m .
- Find the distance between $(2, 1)$ and $(5, 5)$.
- Find the midpoint of the segment joining $(4, 8)$ and $(10, 2)$.
- Find the perpendicular distance from the origin to the line $3x + 4y - 10 = 0$.
- A line has x -intercept 4 and y -intercept 6. Find its gradient.
- Find the area of the triangle with vertices $(0, 0)$, $(4, 0)$ and $(0, 3)$.
- The points $(1, 2)$, $(2, 4)$ and $(k, 10)$ are collinear. Find k .
- Using the shoelace formula, find the area of the triangle with vertices $(0, 0)$, $(3, 4)$ and $(6, 0)$.

Section B Manipulation Drills

(≤50 s each)

Build the line, then interrogate it. Extra tool required per question.

- The line $2x - 3y + 6 = 0$ is perpendicular to which of the following?

A) $y = -\frac{3}{2}x + 1$

B) $y = \frac{3}{2}x - 2$

C) $y = \frac{2}{3}x + 5$

D) $y = -\frac{2}{3}x + 3$

2. What is the perpendicular distance from $(2, 3)$ to the line $5x + 12y - 7 = 0$?
- A) 1
B) 2
C) 3
D) 4
3. Which point lies on the perpendicular bisector of the segment joining $(1, 2)$ and $(5, 6)$?
- A) $(2, 5)$
B) $(3, 3)$
C) $(4, 4)$
D) $(5, 3)$
4. Find the area of the quadrilateral with vertices $(0, 0)$, $(4, 0)$, $(5, 3)$, $(1, 3)$, in that order.
- A) 10
B) 12
C) 14
D) 16
5. The line through $(1, 2)$ and $(4, 5)$ crosses the y -axis at which value?
- A) 0
B) 1
C) 2
D) 3
6. Find the perpendicular distance from $(0, 1)$ to the line $6x - 8y + 2 = 0$.
7. Which of the following is NOT parallel to $y = 3x + 1$?
- A) $y = 3x + 5$
B) $6x - 2y = 4$
C) $3x - y = 2$
D) $y = 2x + 1$
8. The line $y = x + 1$ meets the line through $(0, 3)$ perpendicular to it at the point P . Find the coordinates of P .
9. Which point is closest to the origin?

- A) (1, 3)
- B) (2, 2)
- C) (4, 0)
- D) (0, 5)

10. Using the shoelace formula, find the area of the quadrilateral with vertices (1, 0), (5, 2), (4, 6), (0, 3), in that order.

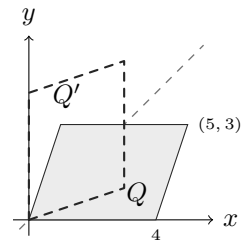
Section C Substitution & Structure

(≤90 s each)

Spot the tool first — the numbers were chosen so that a clean route exists. These are genuinely harder: non-obvious structure, hidden invariants, and competition traps.

1. The line $l : y = 6 - 2x$ meets the x -axis at A . A line m , perpendicular to l , meets the negative x -axis at $(k, 0)$. The lines l , m and the x -axis enclose a triangle of area 20. What is k ? (after TMUA 2017 Paper 1 Q3)
 - A) -7
 - B) -10
 - C) $3 - 5\sqrt{2}$
 - D) $3 - 2\sqrt{5}$
2. The perpendicular bisector of the segment joining $A(2, -6)$ and $B(b, 4)$, where $b > 0$, passes through the origin. What is b ? (after TMUA Specimen Paper 1 Q3)
 - A) $2\sqrt{10}$
 - B) $2\sqrt{14}$
 - C) 12
 - D) $2\sqrt{6}$
3. The point $A(2, 9)$ is reflected in the line $x + 2y = 5$ to give the point A' . How far is A' from the origin?
 - A) $\sqrt{85}$
 - B) $\sqrt{10}$
 - C) 5
 - D) $\sqrt{185}$
4. The lines $3x - 4y + 2 = 0$ and $4x - 3y - 5 = 0$ cross, dividing the plane into four angular regions. Which line bisects the angle whose region contains the point (1, 1)?
 - A) $x - y = 7$
 - B) $x + y = 7$
 - C) $7x + 7y = 3$
 - D) $7x - 7y = 3$

5. The points $A(1, 4)$, $B(4, 7)$ and $C(10, k)$ are collinear. The line through C perpendicular to AB meets the y -axis at D . What is the y -coordinate of D ?
- A) 23
B) 24
C) 25
D) 26
6. Triangle ABC has vertices $A(0, 0)$, $B(9, 3)$ and $C(3, 9)$. The point P lies on AB with $AP : PB = 1 : 2$, and Q lies on AC with $AQ : QC = 2 : 1$. What is the area of the quadrilateral $PBCQ$?
- A) 24
B) 28
C) 27
D) 32
7. The lines $y = 2x + 1$, $y = 7 - x$ and $y = \frac{1}{2}x - 2$ enclose a triangle. What is its area?
- A) 12
B) 18
C) 24
D) 48
8. The quadrilateral Q has vertices $(0, 0)$, $(4, 0)$, $(5, 3)$ and $(1, 3)$ in that order (a parallelogram). A second quadrilateral Q' is obtained by reflecting Q in the line $y = x$. What is the area of the overlap $Q \cap Q'$?
- A) 6
B) 8
C) 9
D) 12



Section D Challenge

(TMUA / SMC / BMO1 difficulty)

Hardest 5: deep insight, hidden structure, or a construction you must discover. No diagram gives away the trick.

1. The points $A(1, 3)$ and $B(7, 5)$ are fixed, and P moves along the line $y = x$. What is the least possible value of $PA + PB$?
- A) $4\sqrt{2}$
B) $2\sqrt{34}$
C) 10
D) $2\sqrt{10}$

2. The square $MNOP$ has side 10. Points R, S, T, U lie on MN, NO, OP, PM respectively, and $RSTU$ is a rectangle whose sides make 45° with the sides of the square. The area of $RSTU$ is 20. What is the length of a diagonal of $RSTU$? (after TMUA 2023 Paper 1 Q5)
- A) $4\sqrt{15}$
B) $4\sqrt{10}$
C) 10
D) $4\sqrt{5}$
3. What is the area of the region of points (x, y) satisfying both $|x| + 2|y| \leq 6$ and $y \geq x - 2$?
- A) 10
B) 18
C) 26
D) 36
4. How many straight lines through the point $(2, 3)$ form, together with the two coordinate axes, a triangle of area 12?
- A) 1
B) 2
C) 4
D) 3
5. For each t with $0 < t < 4$, the line L_t passes through $(t, 0)$ and $(0, 4 - t)$. What is the greatest possible distance from the origin to L_t ?
- A) 1
B) 2
C) $\sqrt{2}$
D) $2\sqrt{2}$

“Shoelace before altitude: area is a one-line cross product, not a construction.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths